

Rotating Unbalance and More 2 DOF Systems

EGR 345

GVSU

Plan for today

- quick review
- quiz 1 and exam 1 plans and topics
- rotating unbalance
- more on 2 DOF systems
 - how can Python help us understand 2 DOF systems?

Quiz 1

→ 10/8 @ 9:45

→ see topics
on github

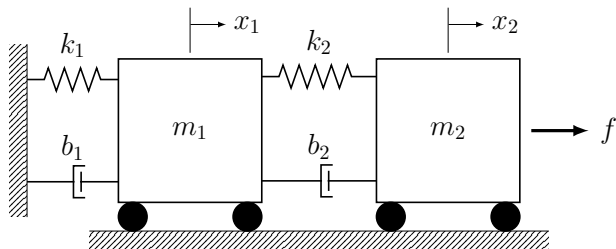
→ Laplace table
provided

→ calculator
allowed

→ No Note sheet

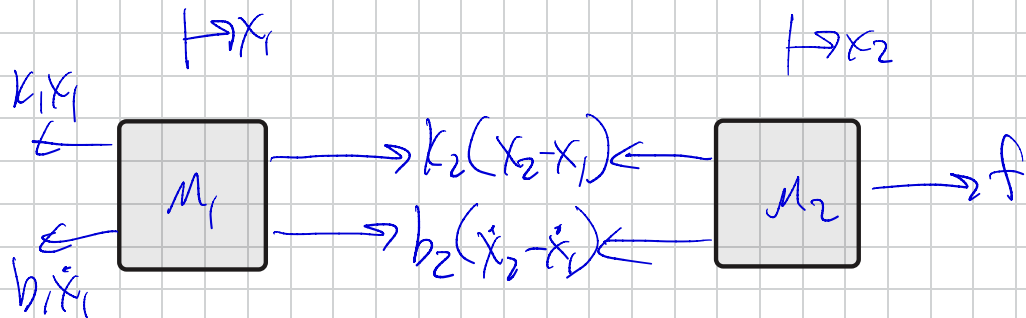
→ 1-2 questions

Review: 2DOF TFs



$$G_1(s) = \frac{?}{?}$$

$$G_2(s) = \frac{?}{?}$$



$$m_1 \ddot{x}_1 = -k_1 x_1 - b_1 \dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1)$$

$$m_1 \ddot{x}_1 + (b_1 + b_2) \dot{x}_1 + (k_1 + k_2) x_1 - b_2 \dot{x}_2 - k_2 x_2 = 0$$

$$\mathcal{D}_1 \left[m_1 s^2 + (b_1 + b_2) s + k_1 + k_2 \right] x_1 - (b_2 s + k_2) x_2 = 0$$

$$M_2 \ddot{x}_2 = f - k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1)$$

$$M_2 \ddot{x}_2 + b_2 \dot{x}_2 + k_2 x_2 - k_2 x_1 - b_2 \dot{x}_1 = f$$

$$\underbrace{(M_2 s^2 + b_2 s + k_2)}_{D_2} x_2 - \underbrace{(b_2 s + k_2)}_N x_1 = 0$$

$$D_1 X_1 - N X_2 = 0 \rightarrow X_1 = \frac{N}{D_1} X_2$$

$$D_2 X_2 - N X_1 = F$$

$$\frac{X_1}{2} = \frac{N}{D_1}$$

$$D_1 \left(D_2 X_2 - \frac{N^2}{D_1} X_2 = F \right) \rightarrow (D_1 D_2 - N^2) X_2 = D_1 F$$

$$\frac{X_2}{F} = \frac{D_1}{D_1 D_2 - N^2} = G_2$$

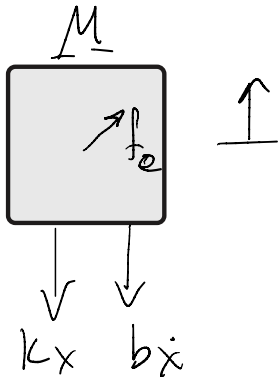
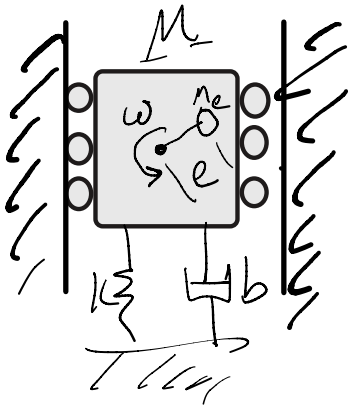
$$\cancel{\frac{x_2}{f} \cdot \frac{x_1}{x_2}} = \frac{x_1}{f} = G_1 = \frac{\cancel{D_1}}{D_1 D_2 N^2} \cdot \frac{N}{\cancel{D_1}}$$

$$G_1 = \frac{N}{D_1 D_2 - N^2} \leftarrow \text{4th order}$$

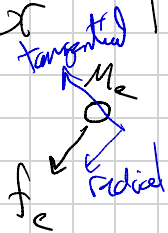
$$G_2 = \frac{D_1}{D_1 D_2 - N^2}$$

Rotating Unbalance

- classic problem that might be used in your project designs



EOM for little / eccentric mass



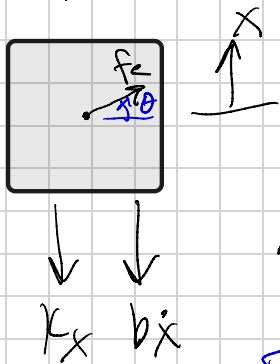
radial acc: $\frac{v^2}{r}$

$\Sigma F_{\text{radial}}: f_c = M_e a_r = M_e \frac{v^2}{r} = M_e \omega^2 r$

$v = \omega r \rightarrow v^2 = \omega^2 r^2$

$f_c = M_e a_r = M_e v^2 / r$

$$f_c = m_e \omega^2 r$$



$$\Sigma f_x: m\ddot{x} = -kx - b\dot{x} + f_c \sin \theta$$

$$\theta = \omega t$$

$$m\ddot{x} + b\dot{x} + kx = m_e \omega^2 r \sin(\omega t)$$

or Idea(s):

- rotating unbalance can cause vibrations
- could also be used for actuation

2 DOF TFs

- they are complicated
- how can we understand them?
- I don't want to do really long PFE problems
 - can Python help us?